Retail Store Sales Analysis

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Course: Data Fundamentals with Data Warehousing, SQL, Python

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1. Introduction

Retail businesses generate large volumes of sales data daily. Analysing this data helps organizations understand customer behaviour, identify top-performing products, and make informed business decisions.

This project focuses on analysing retail sales data of electronic products such as laptops, mobiles, headphones, and keyboards across multiple cities including Pune, Mumbai, and Nagpur.

The objective is to extract meaningful insights using tools like Excel, Python, and SQL.

2. Problem Statement

A retail company has observed fluctuations in its sales but lacks clarity on:

- Which products generate the highest revenue
- Which cities contribute the most to sales
- Which product categories perform best
- How sales change over time

The company requires a data-driven approach to identify patterns and improve its marketing and sales strategies.

3. Objectives

The main objectives of this project are:

- To calculate total revenue generated from sales
- To identify the best-selling product
- To analyze city-wise sales performance
- To visualize sales trends
- To perform analysis using multiple tools (Excel, Python, SQL)

4. Dataset Description

The dataset used in this project is a CSV file containing sales records.

Attributes:

- **Order ID** – Unique identifier for each order
- **Order Date** – Date of purchase
- **City** – Location of the store
- **Product** – Name of the product

- **Category** – Product category (Electronics/Accessories)
- **Quantity** – Number of units sold
- **Price** – Price per unit

A new column **Total Sales** is calculated as:

Total Sales = Quantity × Price

5. Tools and Technologies Used

- **Microsoft Excel** – Data analysis using Pivot Tables
- **Python (Pandas, Matplotlib)** – Data processing and visualization
- **SQL** – Query-based data analysis

6. Methodology

6.1 Excel Analysis

- Imported dataset into Excel
- Created a new column: Total Sales
- Generated Pivot Table:
  - Rows → Product
  - Columns → City
  - Values → Sum of Total Sales

Sum of Total Sales	city			
product	Mumbai	Nagpur	Pune	Grand Total
Headphones			10000	10000
Keyboard		6000		6000
Laptop	50000		100000	150000
Mobile	60000			60000
Grand Total	**110000**	**6000**	**110000**	**226000**

Figure 1: Pivot Table showing product vs city sales

- Identified:
  - Best-performing product
  - Highest revenue-generating city

6.2 Python Analysis

- Loaded dataset using Pandas

- Checked data structure using .info() and .describe()
- Created a new column for total sales
- Performed:
    - Total revenue calculation
    - Product-wise sales analysis
    - City-wise revenue analysis

```
Matplotlib is building the font cache; this may take a moment.
   order_id  order_date    city  ...     category quantity  price
0      1001  2025-01-10    Pune  ...  Electronics        2  50000
1      1002  2025-01-11  Mumbai  ...  Electronics        3  20000
2      1003  2025-01-11    Pune  ...  Accessories        5   2000
3      1004  2025-01-12  Nagpur  ...  Accessories        4   1500
4      1005  2025-01-13  Mumbai  ...  Electronics        1  50000

[5 rows x 7 columns]
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   order_id    5 non-null      int64
 1   order_date  5 non-null      str
 2   city        5 non-null      str
 3   product     5 non-null      str
 4   category    5 non-null      str
 5   quantity    5 non-null      int64
 6   price       5 non-null      int64
dtypes: int64(3), str(4)
memory usage: 412.0 bytes
None
          order_id  quantity        price
count     5.000000  5.000000      5.00000
mean   1003.000000  3.000000  24700.00000
std       1.581139  1.581139  24268.29207
min    1001.000000  1.000000   1500.00000
25%    1002.000000  2.000000   2000.00000
50%    1003.000000  3.000000  20000.00000
75%    1004.000000  4.000000  50000.00000
max    1005.000000  5.000000  50000.00000
Total Revenue: 226000
product
Headphones    5
Keyboard      4
Laptop        3
Mobile        3
Name: quantity, dtype: int64
city
Mumbai    110000
Nagpur      6000
Pune      110000
Name: total_sales, dtype: int64
```

Figure 2: Python output displaying dataset and total revenue

6.3 SQL Analysis

Performed queries to:

- Display all records
- Calculate total revenue
- Identify top-selling product
- Compute city-wise revenue
- Analyse category-wise performance
- Find highest-value order
- Calculate average product price

The SQL query results were consistent with the findings obtained from Excel and Python analysis.

7. Data Visualization

The following visualizations were created:

- **Bar Chart** → Product vs Total Sales

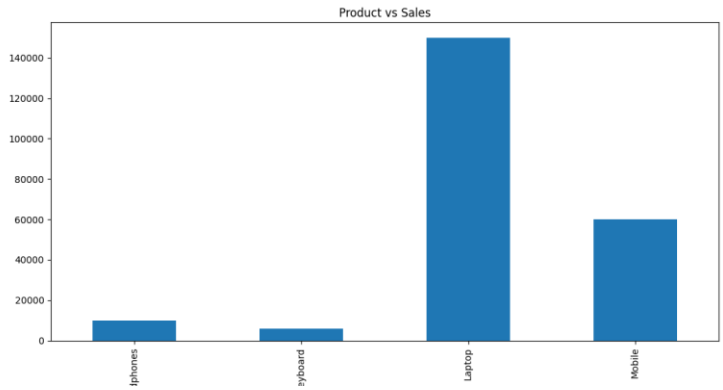

Figure 3: Bar chart showing product-wise sales

- **Pie Chart** → Category-wise sales distribution

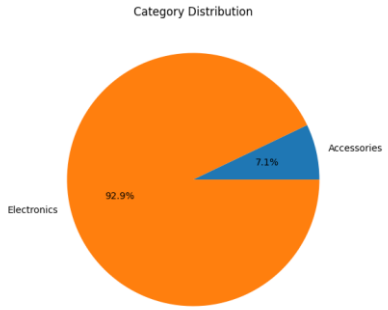

Figure 4: Pie chart showing category distribution

- **Line Chart** → Sales trends over time

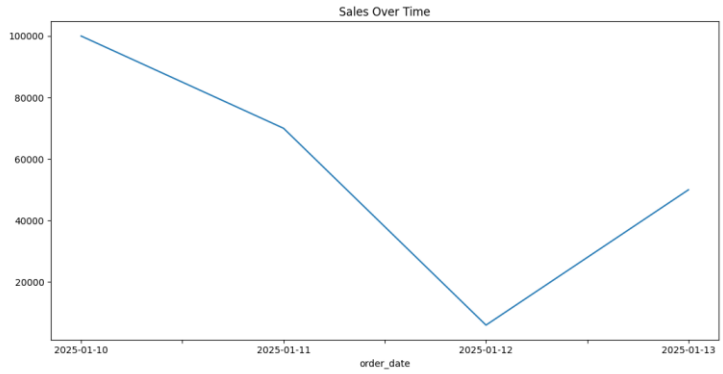

Figure 5: Line chart showing sales trends over time

These visualizations help in better understanding patterns and comparisons.

8. Results and Findings

- The **Laptop** is the best-performing product with the highest total revenue of ₹150000
- **Mumbai and Pune** are the top-performing cities, each generating ₹110000 in revenue
- Accessories contribute less revenue compared to electronics
- Sales are concentrated in major urban areas

9. Unique Features

- Integration of multiple tools (Excel, Python, SQL)
- Clear and structured data analysis approach
- Use of visualizations for better understanding
- Simple and scalable design

10. Future Improvements

- Develop interactive dashboards using Power BI or Tableau
- Incorporate real-time data analysis
- Apply machine learning for sales prediction
- Expand dataset for deeper insights

11. Conclusion

This project successfully demonstrates how sales data can be analyzed to extract meaningful business insights. By using Excel, Python, and SQL, important patterns such as top products and high-performing cities were identified.

The findings can help businesses optimize inventory, improve marketing strategies, and make data-driven decisions.